



General purpose PLL

2026 July 9

Super Long Wave

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Fatima

Agenda

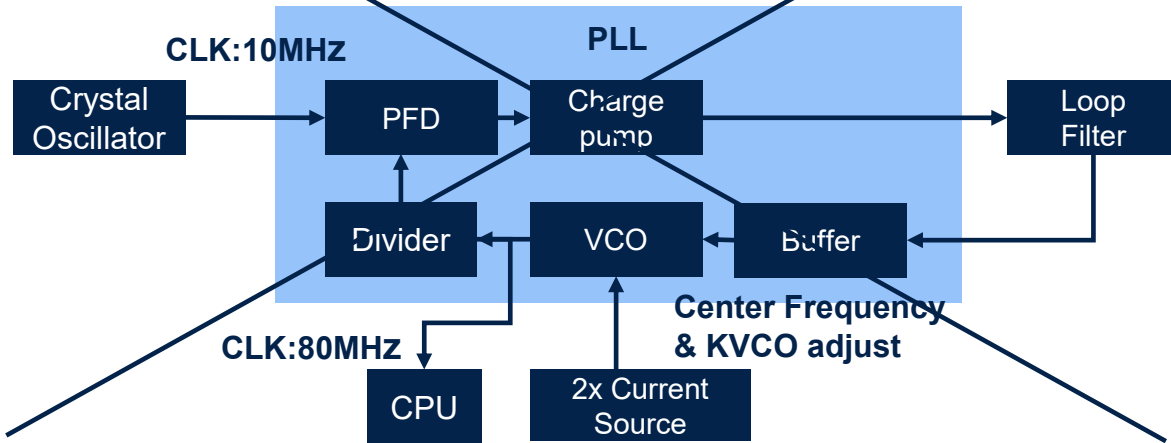
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Project information

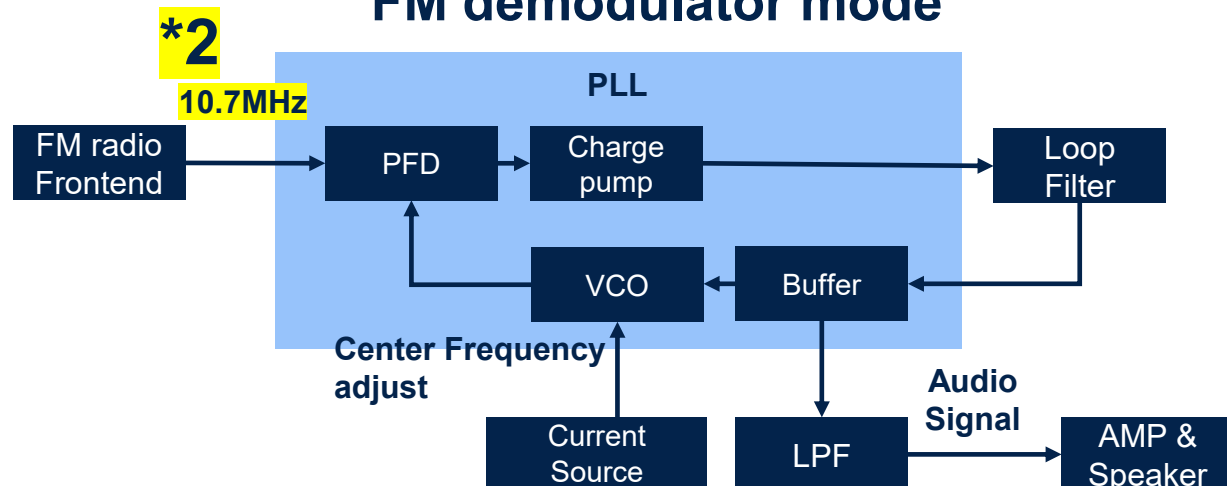
- Project name – General purpose PLL
- Team members
 - Yutaka Kotani (team leader)
 - @Yutaka KOTANI
 - Ashutosh Chakravarty - @Teddy
 - Cheng Fei - @Buttercutter
 - Fatima - @fa_84
- Review date: 2026 July 9
- Review version: git tag: v0.1
- Track C

Design objective

Clock multiplier mode



FM demodulator mode



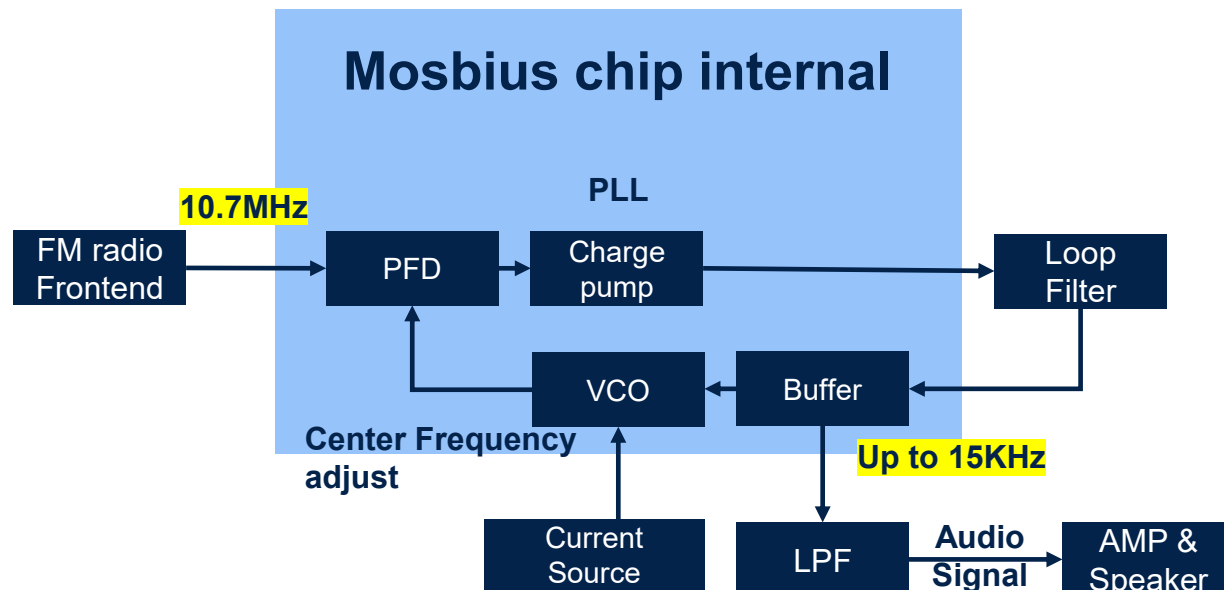
- What problems circuit solves
 - The PLL extracts the audio signal by performing FM demodulation on the 10.7 MHz intermediate frequency (IF) signal from the preceding radio front-end.
- Target specifications
 - Input FM modulated IF 10.7MHz
 - Output FM demodulated audio signal
- Intended Use
 - Create FM radio demodulator
- Project scope and spec changes

Following our mentor's advice, we have reduced the project scope and specifications.

 1. Clock multiplier mode was deleted
 2. Input signal was changed from a direct FM signal to an intermediate frequency of 10.7 MHz

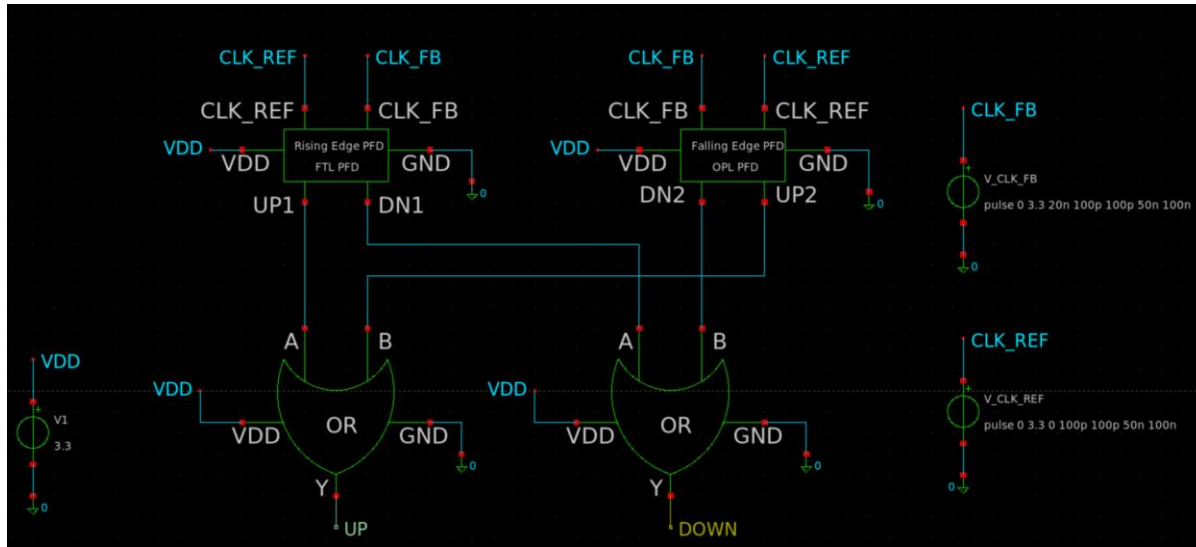
System overview

- PFD
 - Compare phase between input FM radio frontend signal and VCO output
 - Generates UP/DOWN signal to charge pump
- Charge pump
 - Converts PFD UP/DOWN signals into source/sink currents.
 - Generates the VCO control voltage through the loop filter while reducing current mismatch and reference spurs.



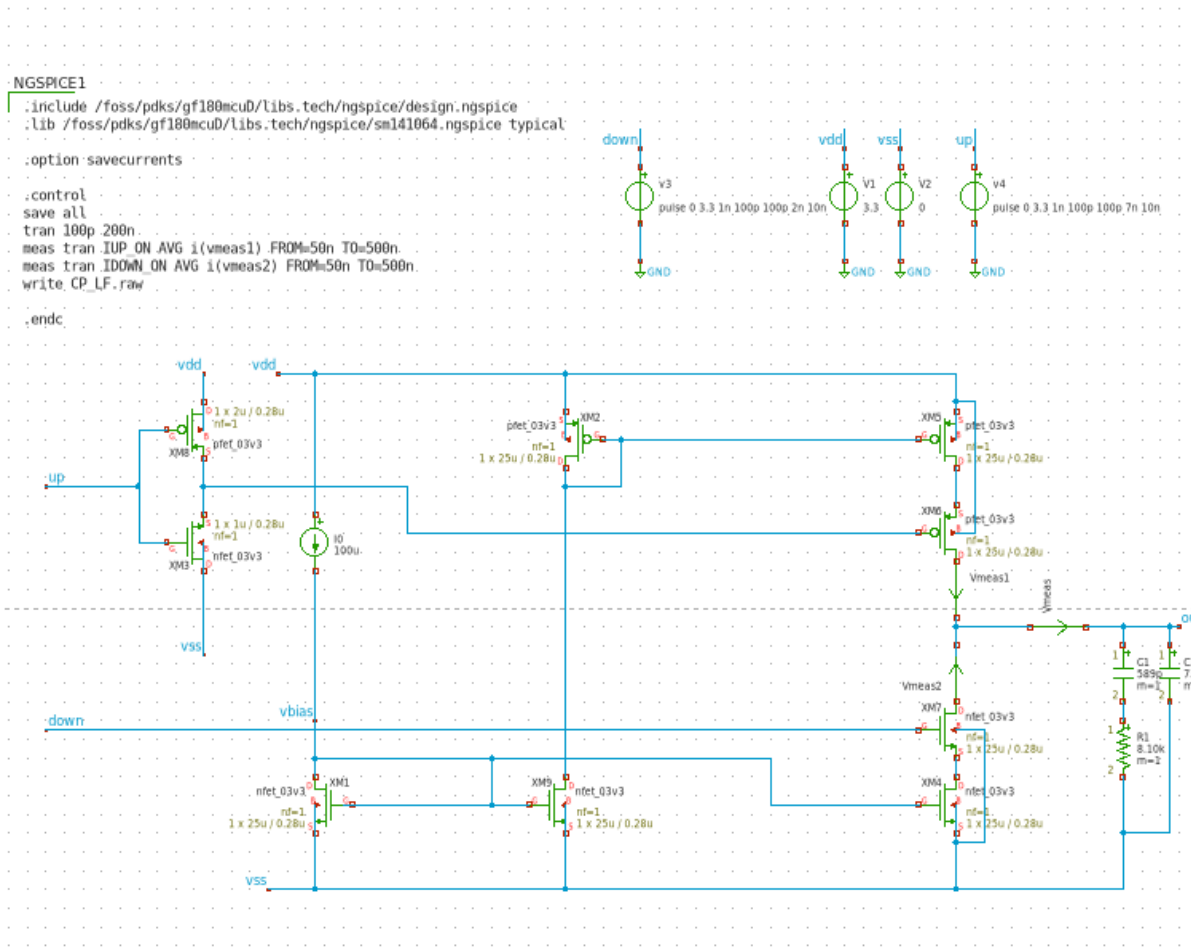
- Loop filter (external)
 - Filters the charge pump current to produce a stable VCTRL.
 - Controls loop bandwidth, stability, and lock time while reducing ripple.
- Buffer
 - To copy the voltage from the loop filter to the VCO. This ensures a reliable input
- LPF (external)
 - For removing high frequency components from buffer for smoother output
- VCO
 - Generate 0-3.3V clock signal according to input signal amplitudes
- Current source (external)
 - Provide adjustable current to set center frequency of VCO

Schematic summary - PFD



- Purpose
 - Compare phase between input FM radio frontend signal and VCO output
 - Generates UP/DOWN signal to charge pump
- Input and Output
 - Input: FM radio frontend signal and VCO output
 - Output: UP/DOWN signal
- Important design decisions
 - T.B.D
- Interface to other blocks
 - Receives rf modulated signal and generate up/down signal to charge pump

Schematic summary – Charge pump/ Loop filter

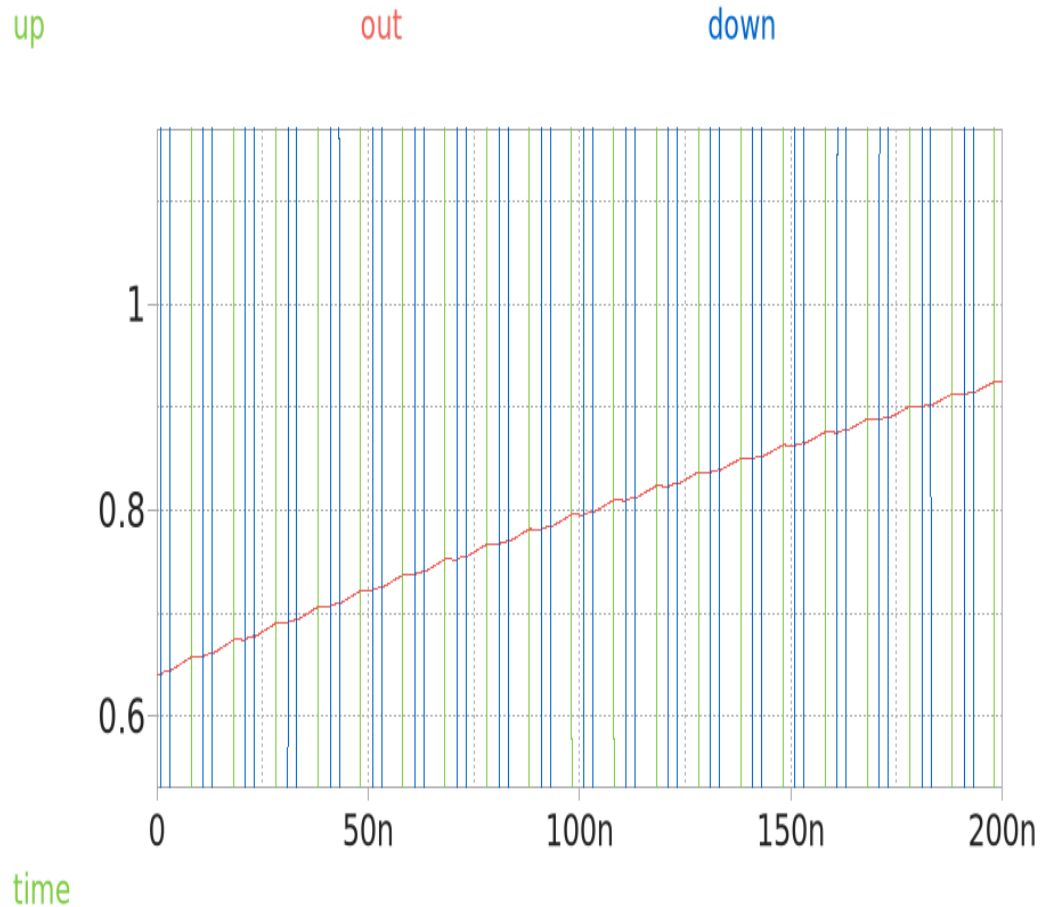


- Purpose
 - Converts PFD UP/DOWN pulses into source/sink currents that charge or discharge the loop filter, generating VCTRL for the VCO
- Input and Output
 - Input: up / down digital control signals from PFD
 - Output: charge/discharge current at “out” node, into loop filter (C1 / R1 / C2)
- Important design decisions
 - Cascaded PMOS/NMOS switch pairs (XM2/XM5/XM6 and XM7/XM4) for high output impedance
 - I_UP / I_DOWN current matching monitored directly in simulation to limit reference spurs
 - 100 μ A reference current (I0) distributed via vbias to all mirror branches
- Interface to other blocks
 - Receives up/down directly from PFD; drives the external loop filter node that sets VCO control voltage



Schematic summary – Charge pump

Transient simulation results

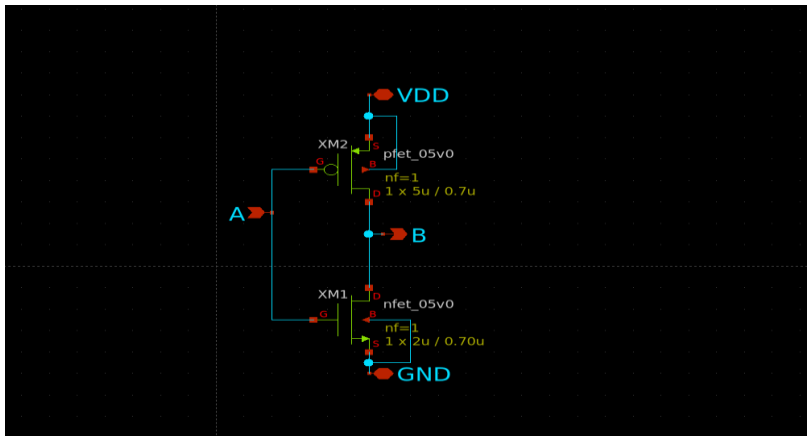
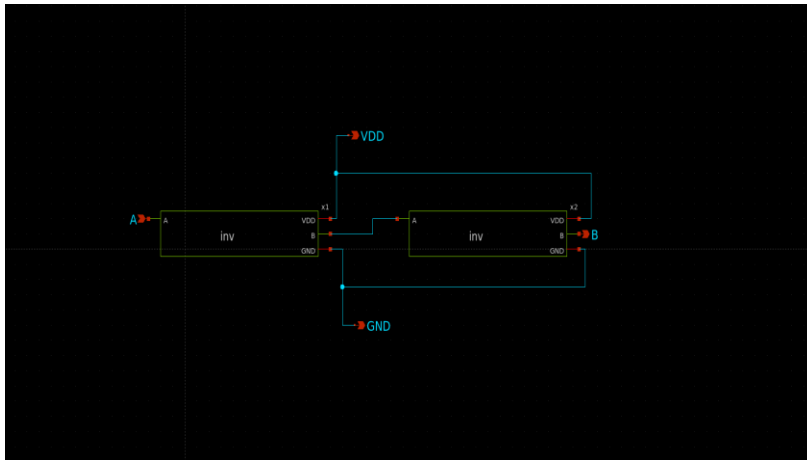


- Testbench
 - ngspice transient, GF180MCU 3.3V devices, up/down pulses (100ps edges) driving charge pump into C1/R1/C2 loop filter
 - IUP_ON_AVG and IDOWN_ON_AVG measured over 50ns–500ns to quantify current matching
- Observed behavior
 - Out node ramps smoothly from ~0.6V toward ~1V as up/down pulses accumulate charge on the loop filter
 - No visible glitching or discontinuity at each switching edge, consistent with well-matched cascode switching
- Next steps
 - Extract numerical I_UP/I_DOWN mismatch and reference-spur estimate across PVT corners



Schematic summary – Buffer

- Schematic (Buffer, Inverter respectively)



- Purpose

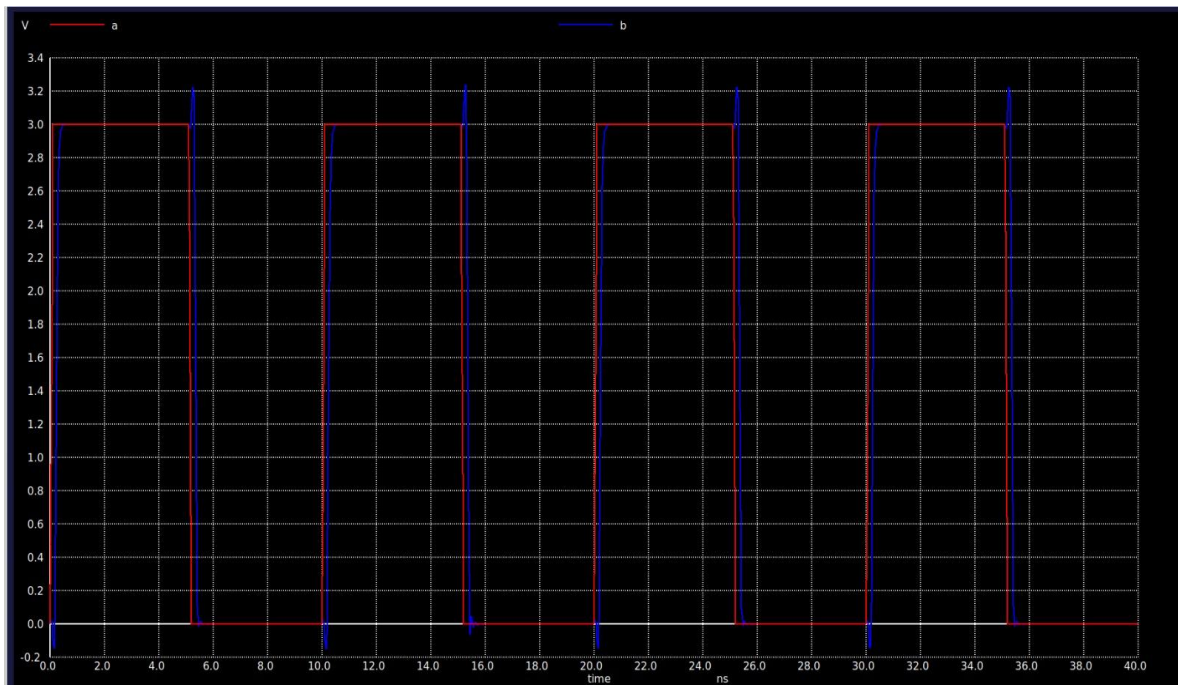
To copy the voltage from the loop filter to the VCO. This ensures a reliable input

- Input and Output

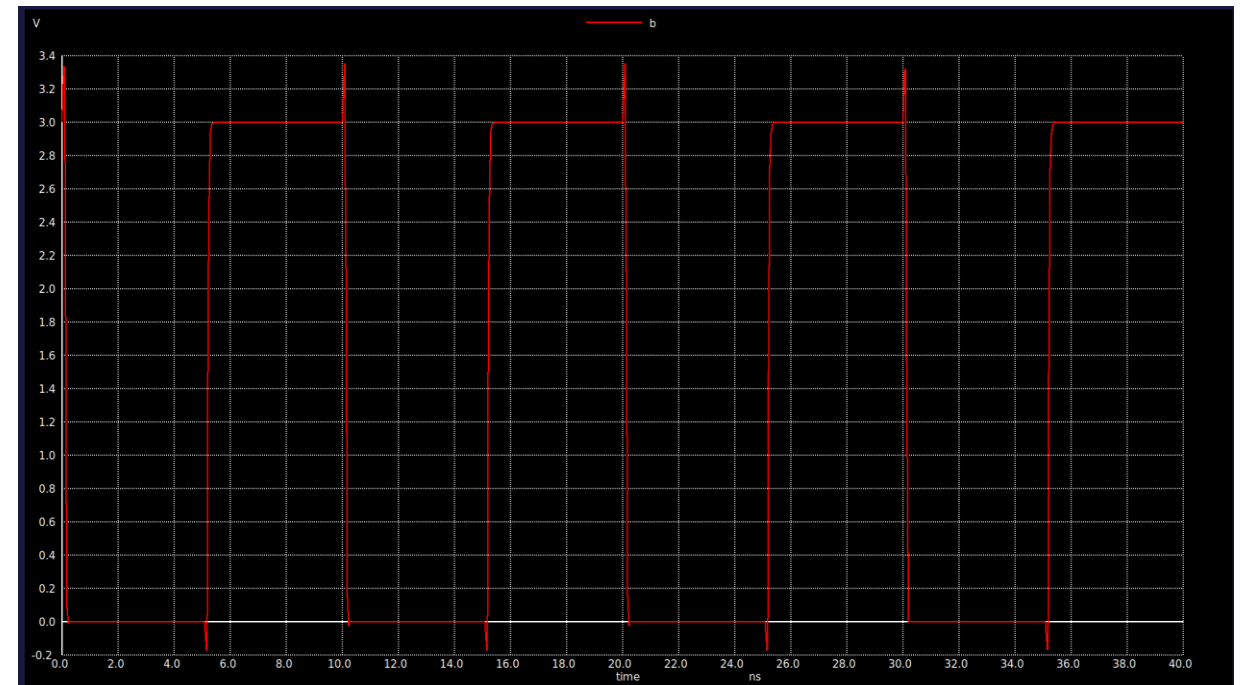
- Input 3V pulse signal (for simulation)
- Output 3V pulse signal identical to input (simulation result)
- 3V VDD input

Simulations of Buffer

- Output of the Buffer

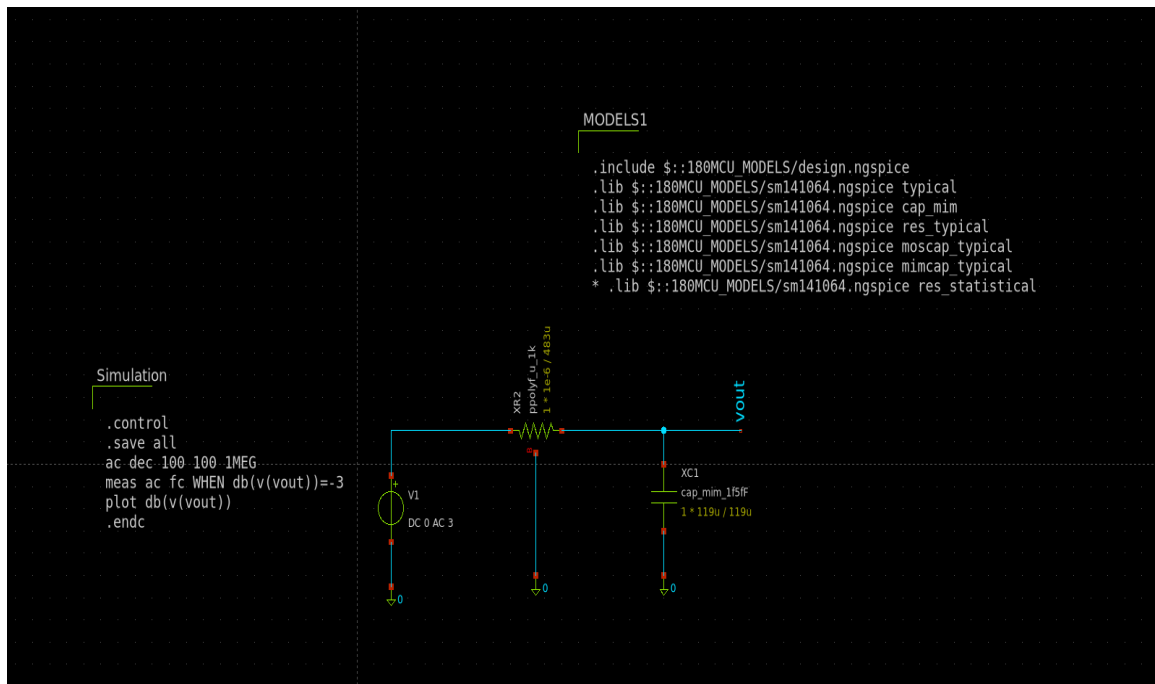


- Output Of Inverter



Schematic summary – LPF

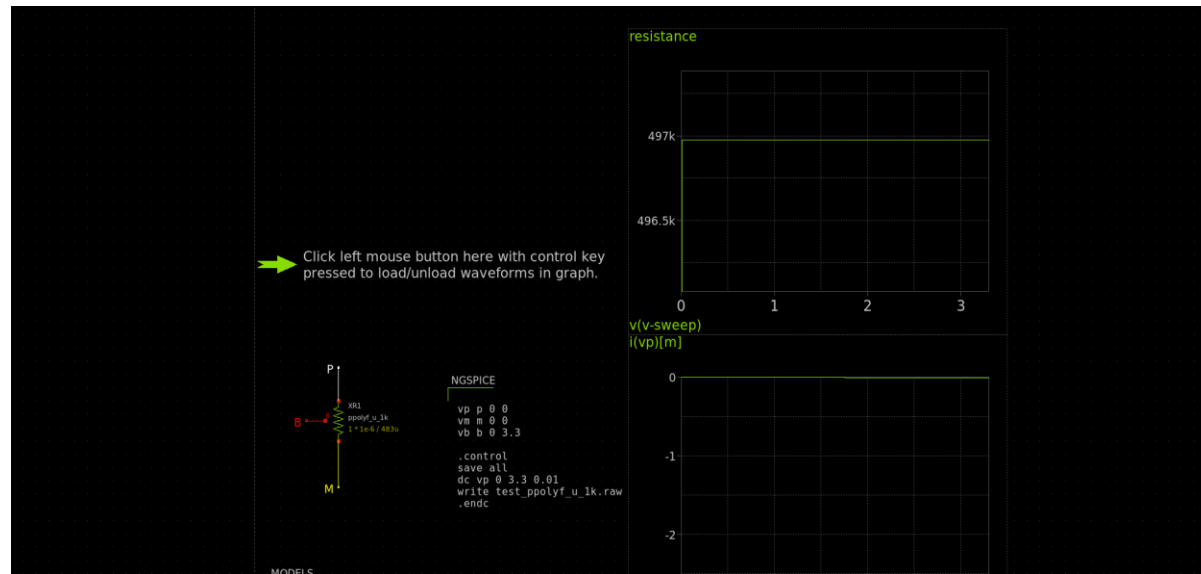
- Schematic



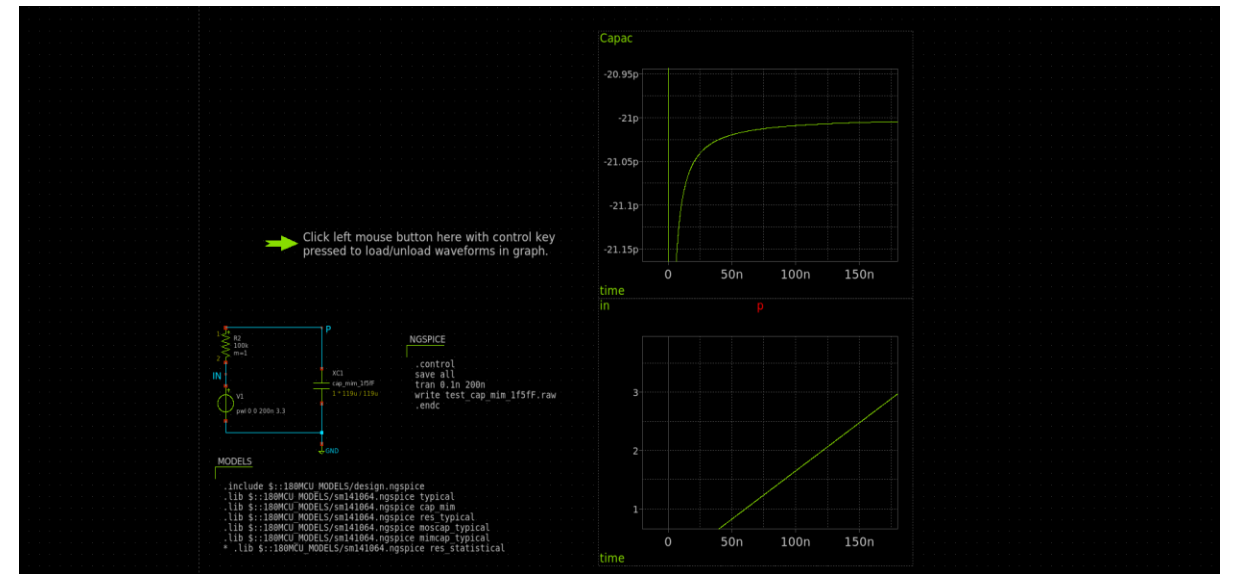
- Purpose
For removing high frequency components from buffer for smoother output
- Input and Output
 - Buffered signal containing high frequency and noise
 - Clean filtered signal
- Important design decisions: Selection of R and C values of reasonable size for filter design
- Interface to other blocks Input from buffer

Resistor and Capacitor Value Selection

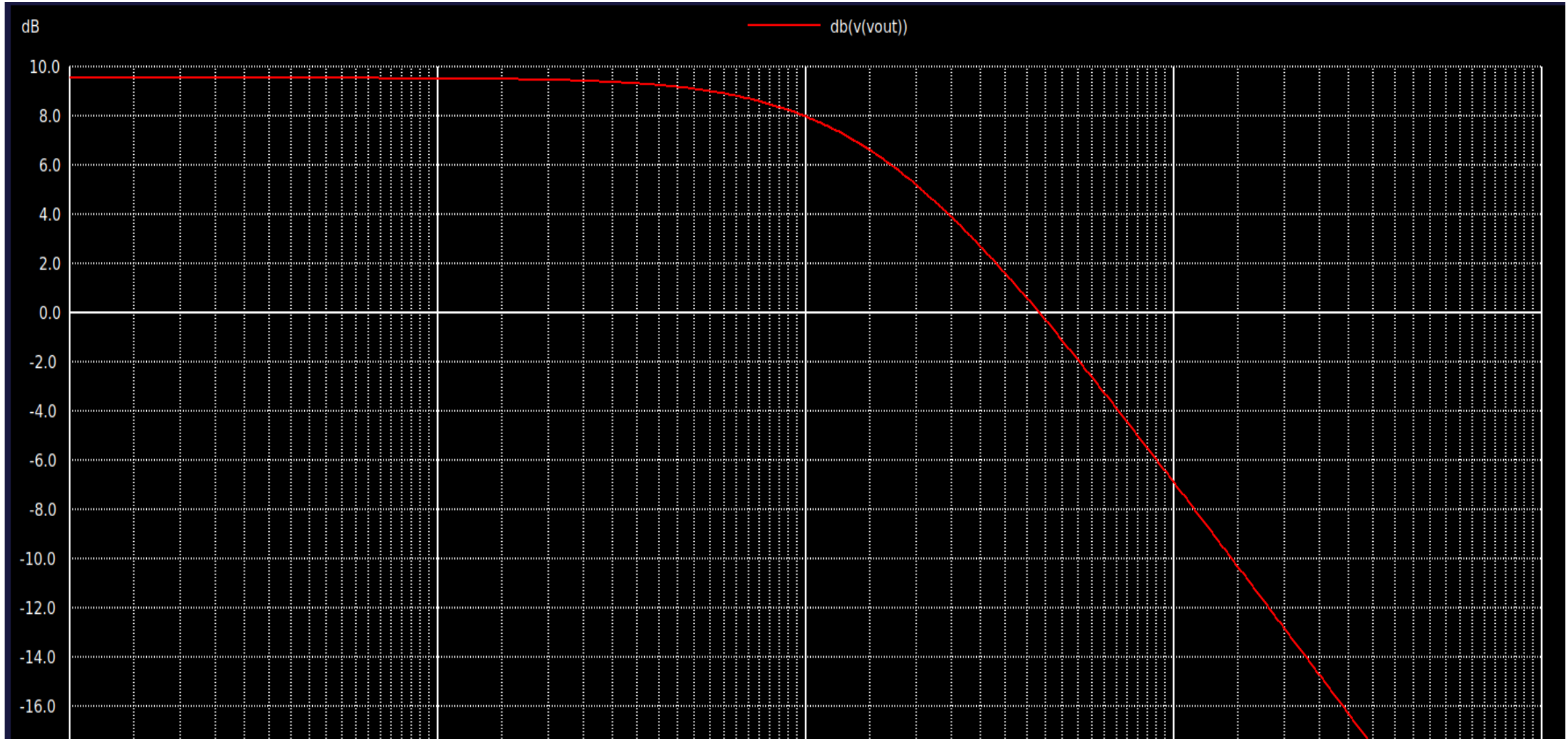
- Resistor Selection



- Capacitor Selection

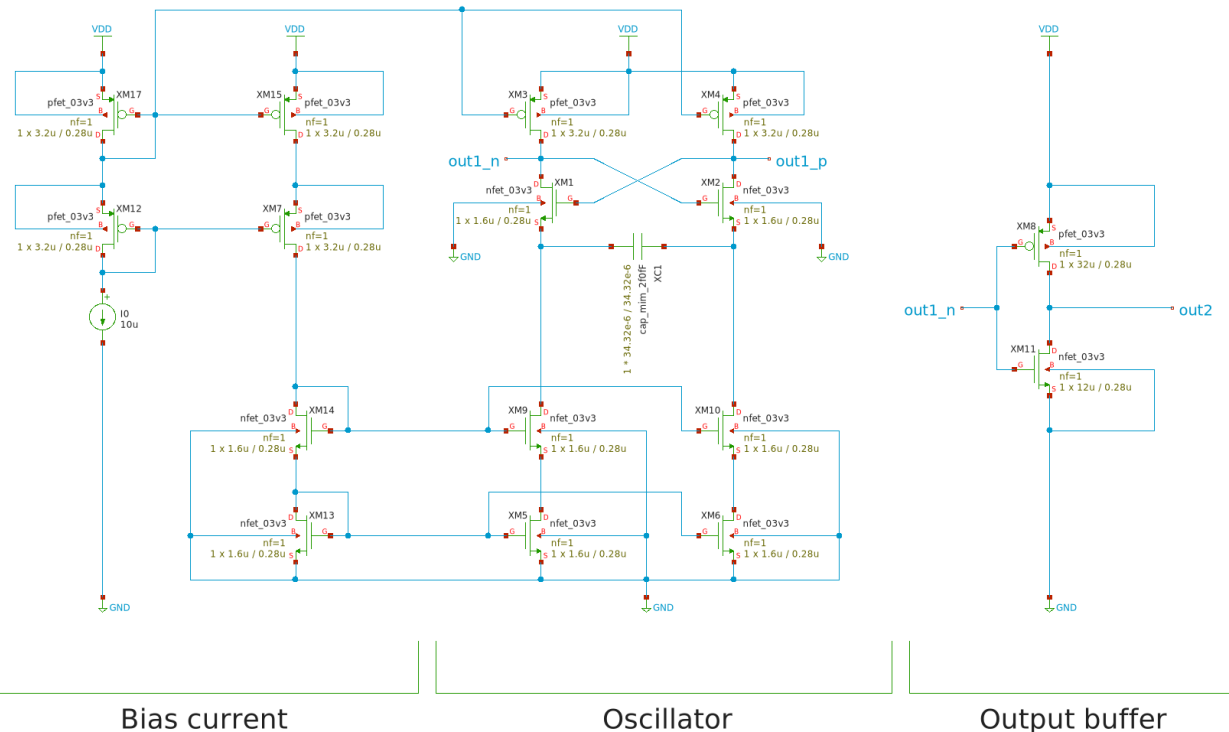


Simulation of LPF



Schematic summary – VCO

- Purpose
 - Generate 0-3.3V clock signal according to input signal amplitudes.
- Input and Output
 - Input: voltage signals that indicate output frequency
 - Output: 10.7MHz $\pm$ 200Khz clock signal
- Important design decisions
 - Define suitable KVCO according to FM Radio spec
 - Assumption: FM radio audio frequency deviation: 15KHz, Input signal max amplitude: **0.2V**
 - Suitable KVCO: **375KHz/V**
 - Using just only one capacitor to tune output frequency for ease of turning
- Interface to other blocks
 - Received VCO control signal from loopfilter; Generate clock signals.



Schematic summary – current source

- schematic diagram T.B.D
- Purpose
 - Provide adjustable current to set center frequency of VCO
- Input and Output
 - Input: tunable resistor with turning knob
 - Output: generate adjustable current up to 200uA
- Important design decisions
 - Fine turning is necessary
- Interface to other blocks
 - Generate reference current to VCO.

Design assumption

Target Spec	Min	Typ	Max
Pin count	8		
Estimated Area	***um2		
pll_in voltage [V]	0.8	-	2.5
pll_out voltage [V]	0		vdd
FM demodulator mode	Min	Typ	Max
pll_in frequency [MHz]	10.6	10.7	10.8
dmod_out voltage [V]	0	-	vdd
Cen_freq_cur [uA]	10	-	110
PVT	Min	Typ	Max
Process	SS	TT	FF
Vdd voltage [V]	3	3.3	3.6
Temperature [degC]	20	25	30

Pin	Pin name	Notes
1	VDD	3.3V
2	GND	
3	PLL_IN	Reference clk input
4	PLL_OUT	VCO output
5	CEN_FREQ_CUR	Center frequency bias current
6	KVCO_ADJ_CUR	kvco adjust bias current
7	LF_IN	External Loop filter input
8	LF_OUT	External Loop filter output

Expected load	Min	Typ	Max
Load impedance [ohm]	48k	100k	330k
Load capacitance [pF]	10	22	33

Verification status

- See “project_status_tracker_super_long_wave.xlsx”

No	Main Component	Sub component	Owner	Status	Schematics	Functional simulation	Simulation Link	Design Checklist	Corner simulation	Monte Carlo	ERC	LVS	DRC
1	PFD		Buttercutter	On going	Done	On going	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
2	Charge pump		fa_84	On going	Done	Pass	CP_LF.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
3	Loop filter (external)		fa_84	On going	Done	Pass	CP_LF.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
4	Buffer		Teddy	On going	Done	Pass	tb_buffer.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
5	LPF (external)		Teddy	On going	Done	Pass	tb_lpf.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
6	VCO	CCO	Yutaka KOTANI	On going	Done	Pass	tb_vco_tran.sch	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
7		OTA	Yutaka KOTANI	On going	On going	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
8	Current source (external)		Yutaka KOTANI	On going	On going	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet
9	Integration		Yutaka KOTANI	Not yet	Not yet	Not yet	T.B.U	T.B.U	Not yet	Not yet	Not yet	Not yet	Not yet

Design checklist

- See “design_checklist/design_checklist_0*_*****.xlsx”

 design_checklist_01_pfd.xlsx
 design_checklist_02_Charge_pump.xlsx
 design_checklist_03_Loop_filter_external.xlsx
 design_checklist_04_Buffer.xlsx
 design_checklist_05_LPF_external.xlsx
 design_checklist_06_vco_cco.xlsx
 design_checklist_07_vco_ota.xlsx
 design_checklist_08_current_source_external.xlsx
 design_checklist_09_integration.xlsx

Design checklist						
Main Component	PFD/Charge pump/Loop filter (external)/Buffer/VCO/Current source (external)/Integration					
Sub component						
Type	Analog/Digital/Mixed					
File name	dummy.sch					
Version tag	v0.1					
Designer	Yutaka KOTANI/Teddy/Buttercutter/fa_84					
Reviewer	Yutaka KOTANI/Teddy/Buttercutter/fa_84					
Review date	Jul-26					
	No	Category	Check item	Description	Expected behavior	Observed behavior
	1	Functionality	Requirements implemented			
	2					
	3		Reset behavior verified			
	4		Start up verified			
	5		Power sequencing considered			
	6	Analog	Bias current checked			
	7		Device operating regions verified			
	8		Stability analyzed			
	9		Noise considered			
	10	Digital	Matching requirement considered			
	11		Timing reviewed			
	12		State machine behavior verified			
	13		Clock crossing identified			
	14	Mixed Signal	Reset strategy documented			
	15		Analog/digital interface reviewed			
	16		Isolation strategy documented			
	17	Reliability	Substrate and supply noise considered			
	18		Device voltage limits checked			
	19		Current density reviewed			
	20		Latch-up prevention considered			
	21	Documentation	ESD strategy identified			
	22		Nets clearly named			
	23		Hierarchy consistent			
	24		Parameters documented			
	25		Symbols match implementation			

Open issues

- See “open_issue_super_long_wave.xlsx”

No	Description	Impact	Proposed solution	Owner	Status
1					
2					
3					
4					
5					
6					
7					
8					
9					

Questions for reviewers

- See “questions_for_reviewers_super_long_wave.xlsx”

No	Category	Priority	Question	Questioner	Answer	Answerer	Status
1							
2							
3							
4							
5							
6							
7							
8							
9							